

TABLE 2
MQTT PUBLISH/SUBSCRIBE DETAILS

Component	Type	Action	Topic	Payload	Function
IoT MQTT app	Publish	Relay switch on	Switchsys/relay1	1	Energises relay
IoT MQTT app	Publish	Relay switch off	Switchsys/relay1	0	De-energises relay
IoT MQTT app	Subscribe	—	Switchsys/relay1/status	1	LED indicator turns green
IoT MQTT app	Subscribe	—	Switchsys/relay1/status	0	LED indicator turns grey
ESP-01	Subscribe	—	Switchsys/relay1	1	Turns relay on
ESP-01	Subscribe	—	Switchsys/relay1	0	Turns relay off
ESP-01	Publish	Relay energised	Switchsys/relay1/status	1	Sends on status to app
ESP-01	Publish	Relay de-energised	Switchsys/relay1/status	0	Sends off status to app

System design

In the IoT MQTT Panel app dashboard, there is a single switch to control the relay and an LED indicator to display the relay status.

When Relay1 switch is turned on, the app publishes payload 1 to the topic switchsys/relay1 and sends it to the ESP-01 over the internet via the Switchsys MQTT broker

(broker URL: mqtt.switchsys.in, port: 1883). Since the ESP-01 is subscribed to the same topic (switchsys/relay1), it receives payload 1 and sets GPIO0 HIGH, thereby energising the relay.

After energising the relay, the ESP-01 publishes the relay status to the topic switchsys/relay1/status. This status message is then sent back to the IoT MQTT Panel app through the MQTT broker. In the app, the LED indicator is subscribed to the topic switchsys/relay1/status. Therefore, when payload 1 is received, the LED indicator turns green, indicating that the relay is on.

Similarly, when the relay switch is turned off in the IoT MQTT Panel app, payload 0 is transmitted to the ESP-01 via the topic switchsys/relay1. The ESP-01 then sets GPIO0 LOW, de-energising the relay, and publishes the updated relay status with payload 0 to the topic switchsys/relay1/status. Upon receiving this payload, the LED indicator changes to grey, indicating that the relay is off. Table 2 shows the MQTT topics and payload details used in the project.

Circuit diagram

Fig. 2 shows the circuit diagram of the system. It is built around the ESP-01 Wi-Fi module, the AMS1117-3.3 voltage regulator (5V DC to 3.3V DC), the ULN2003A relay driver IC, the HLK-PM01 230V AC to 5V DC converter module, the HF46F/5-HS1 relay, the 1N4007 diode, and a few other components.

```
// GPIO0 for ESP-01
// NOTE: GPIO0 MUST be HIGH at boot, otherwise ESP-01 will not start
#define Relay1 0
```

Fig. 3: Relay control pin configuration

```
// ===== WIFI CREDENTIALS =====
const char* ssid = "xxxxx"; // WiFi SSID to be updated as per user
const char* password = "xxxxx"; // WiFi Password to be updated by user

WiFi.begin(ssid, password); // Start WiFi connection
```

Fig. 4: Wi-Fi configuration

```
// ===== SWITCHSYS MQTT CONFIG =====
const char* mqttServer = "mqtt.switchsys.in"; // MQTT Broker address

// Unique MQTT client ID (must be unique on broker)
String clientId = "Switchsys_relay1_1234";

#define sub "switchsys/relay1" // Subscribe topic (control)
#define pub "switchsys/relay1/status" // Publish topic (status)
```

Fig. 5: MQTT broker configuration

```
client.setServer(mqttServer, 1883);
```

Fig. 6: MQTT initialisation

```
// If payload is '1' → Relay ON
if ((char)payload[0] == '1') {
    digitalWrite(Relay1, HIGH); // Relay ON
    client.publish(pub, "1"); // Publish ON status
}

// If payload is '0' → Relay OFF
else if ((char)payload[0] == '0') {
    digitalWrite(Relay1, LOW); // Relay OFF
    client.publish(pub, "0"); // Publish OFF status
}
```

Fig. 7: Displays the current relay state

The system uses an ESP-01 Wi-Fi module, a relay module, the Switchsys MQTT broker, and the IoT MQTT Panel app. It becomes operational when powered on. The ESP-01 connects to the Wi-Fi network and communicates with the

MQTT broker. The user sends on/off commands through the mobile app, which the ESP-01 receives via the broker. Based on the received command, the ESP-01 activates or deactivates the relay to switch the connected bulb on or off.